Assignment 3 — Fashion-MNIST Classification

Neural Networks | Individual assignment

Deadline: See course platform. Estimated effort: approximately 2–3 hours.

Objective

Build and compare simple neural networks, and explain how they are trained and evaluated. Focus on understanding the concepts introduced in the lectures rather than achieving the highest accuracy.

Dataset and preparation

Use the Fashion-MNIST dataset (tf.keras.datasets.fashion_mnist.load_data()), containing 28 × 28 grayscale images in 10 clothing categories. Normalize the pixel values to 0–1 and flatten each image into 784 input features (or use a Keras Flatten layer).

Starter code — loading and preparing Fashion-MNIST

```
import tensorflow as tf

# Load Fashion-MNIST
(X_train, y_train), (X_test, y_test) = (
    tf.keras.datasets.fashion_mnist.load_data()
)

# Normalize pixel values to [0, 1]
X_train = X_train.astype("float32") / 255.0
X_test = X_test.astype("float32") / 255.0

# Flatten 28 x 28 images into 784 features
X_train = X_train.reshape(-1, 784)
X_test = X_test.reshape(-1, 784)

# Next step:
# Split the training data into training and validation subsets.
# Keep the test data separate for final evaluation.
```

Use the same training and validation data for all experiments. Train each model for 10 epochs with a batch size of 64. Use sparse categorical cross-entropy as the loss function and a 10-neuron softmax output layer. Use the default optimizer settings in Keras.

Part A — Build and compare neural networks

Task 1. Build a baseline model with one hidden layer of 32 neurons using ReLU activation and the Adam optimizer. Train it and record its training and validation accuracy.

Task 2. Starting from the baseline, run three more experiments. For each experiment, change only the setting shown below; keep the other settings the same as the baseline.

Model	Hidden neurons	Hidden activation	Optimizer
Baseline	32	ReLU	Adam
Experiment 1	64	ReLU	Adam
Experiment 2	32	tanh	Adam
Experiment 3	32	ReLU	SGD

Task 3. Record the training and validation accuracy of all four models in a results table. Identify the model with the highest validation accuracy. Evaluate that model on the test set and report its test accuracy. Do not use the test set to choose the model.

Part B — Explain your neural network

Q1. Architecture and parameters. For the baseline model (784 inputs, 32 hidden neurons and 10 outputs), calculate the number of trainable weights and biases. Explain the difference between a parameter and a hyperparameter.

Q2. Activation and loss. Why do we use ReLU in the hidden layer and softmax in the output layer? Why is sparse categorical cross-entropy appropriate for this dataset?

Q3. Training and Keras functions. Explain the purposes of model.compile(), model.fit(), model.evaluate() and model.predict(). Briefly explain what backpropagation and the optimizer do during training.

Q4. Interpreting your experiments. Which model achieved the highest validation accuracy? Did changing the number of neurons, activation function or optimizer always improve accuracy? Explain using your results.

Keep your answers brief (approximately 3–5 sentences per question, apart from calculations). Use the lecture slides to help you explain the concepts in your own words.

Part C — Understand classification errors

Generate a confusion matrix for your selected model using the test data. Identify one pair of clothing categories the model confuses. Explain in 2–3 sentences what the confusion matrix shows that overall accuracy does not.

Submission

Submit one runnable, clearly labelled Colab/Jupyter notebook containing your code, results table, selected model's test accuracy, confusion matrix and answers to Q1–Q4. No separate report is required. Be prepared to discuss your code and results during the oral examination.

Not required: convolutional neural networks, K-fold cross-validation, manual backpropagation calculations or extensive hyperparameter searches.